

DIFFUSION-BASED GENERATIVE MODELING FOR ACCELERATING NASA GMAO'S ATMOSPHERIC DATA ASSIMILATION SYSTEM

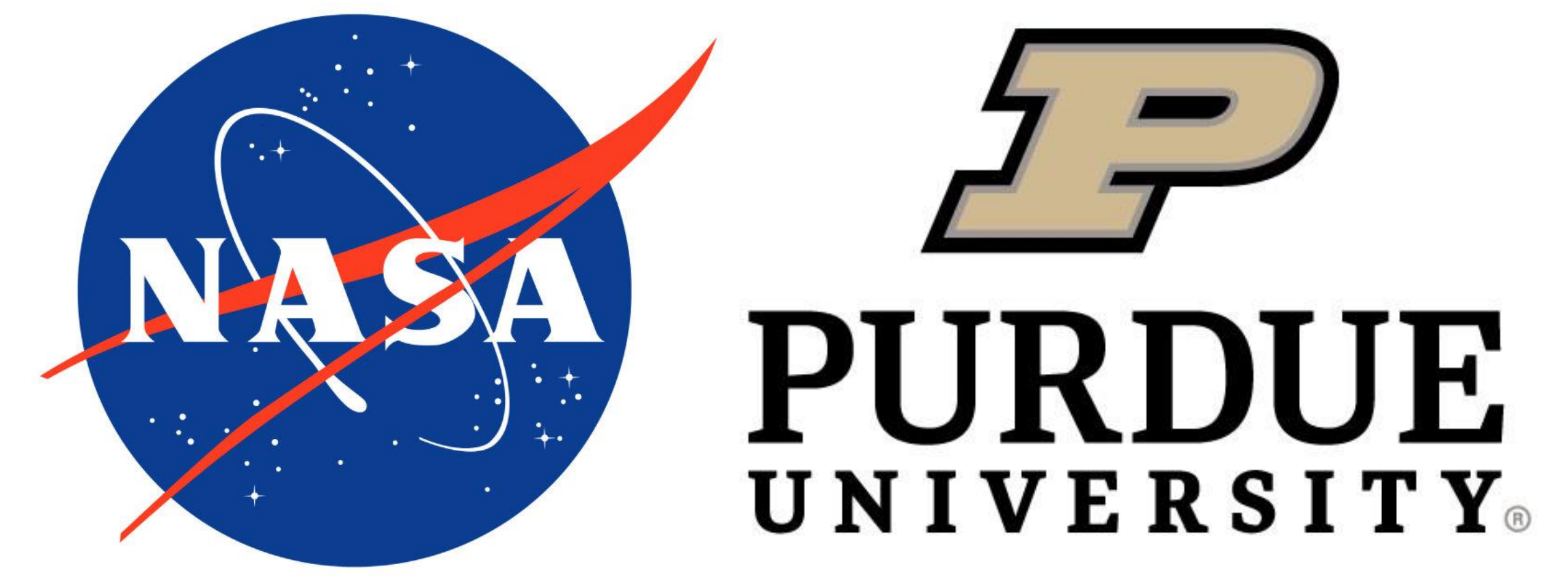
Melissa Adrian¹, Troy Arcomano², Alexander Wikner³, Dibyajyoti Chakraborty⁴, Steven Greybush⁵, Ricardo Todling⁶, Romit Maulik^{1,7}

¹Purdue University, School of Mechanical Engineering ; ²Allen Institute for AI; ³University of Chicago, Department of Geophysical Sciences; ⁴The Pennsylvania State University, College of Information Sciences and Technology;

⁵The Pennsylvania State University, Department of Meteorology and Atmospheric; ⁶NASA, Global Modeling and Assimilation Office; ⁷Argonne National Laboratory, Mathematics and Computer Sciences Division

OVERVIEW

- NASA's hybrid data assimilation system relies on large-scale physics-based equations describing the atmosphere, however, generating ensembles from these equations is **computationally expensive**.
- We aim to build a **surrogate model** that allows for **cheap ensemble generation** via generative diffusion, thereby **accelerating data assimilation**.

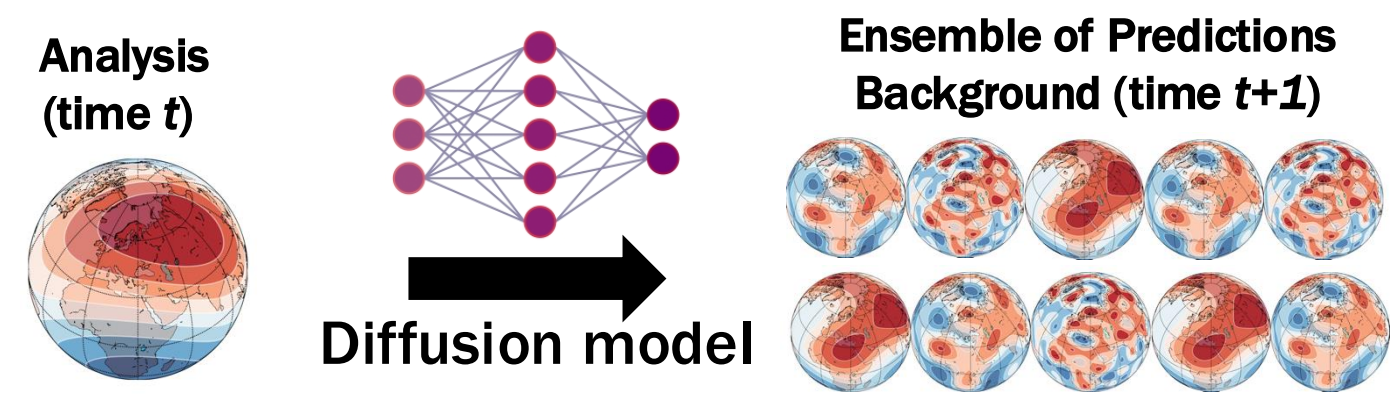


ACKNOWLEDGMENTS

This material is based upon work supported by the National Aeronautics and Space Administration under Grant No. 80NSSC25K0026 issued through the Advanced Information Systems Technology (AIST) Program. This research used resources of the Argonne Leadership Computing Facility, which is a U.S. Department of Energy Office of Science User Facility operated under contract DE-AC02-06CH11357. M.A. acknowledges the support of the Lillian Gilbreth Postdoctoral Fellowship at Purdue University.

MOTIVATION

Atmospheric data assimilation (DA) involves combining multiple sources of information (e.g., observations, forecasts, etc.) to construct an estimate of the state of the atmosphere across time, which involves solving expensive, large-scale, physics-based PDE equations. **We aim to build a generative diffusion-based surrogate of the following mapping in NASA's DA system:**

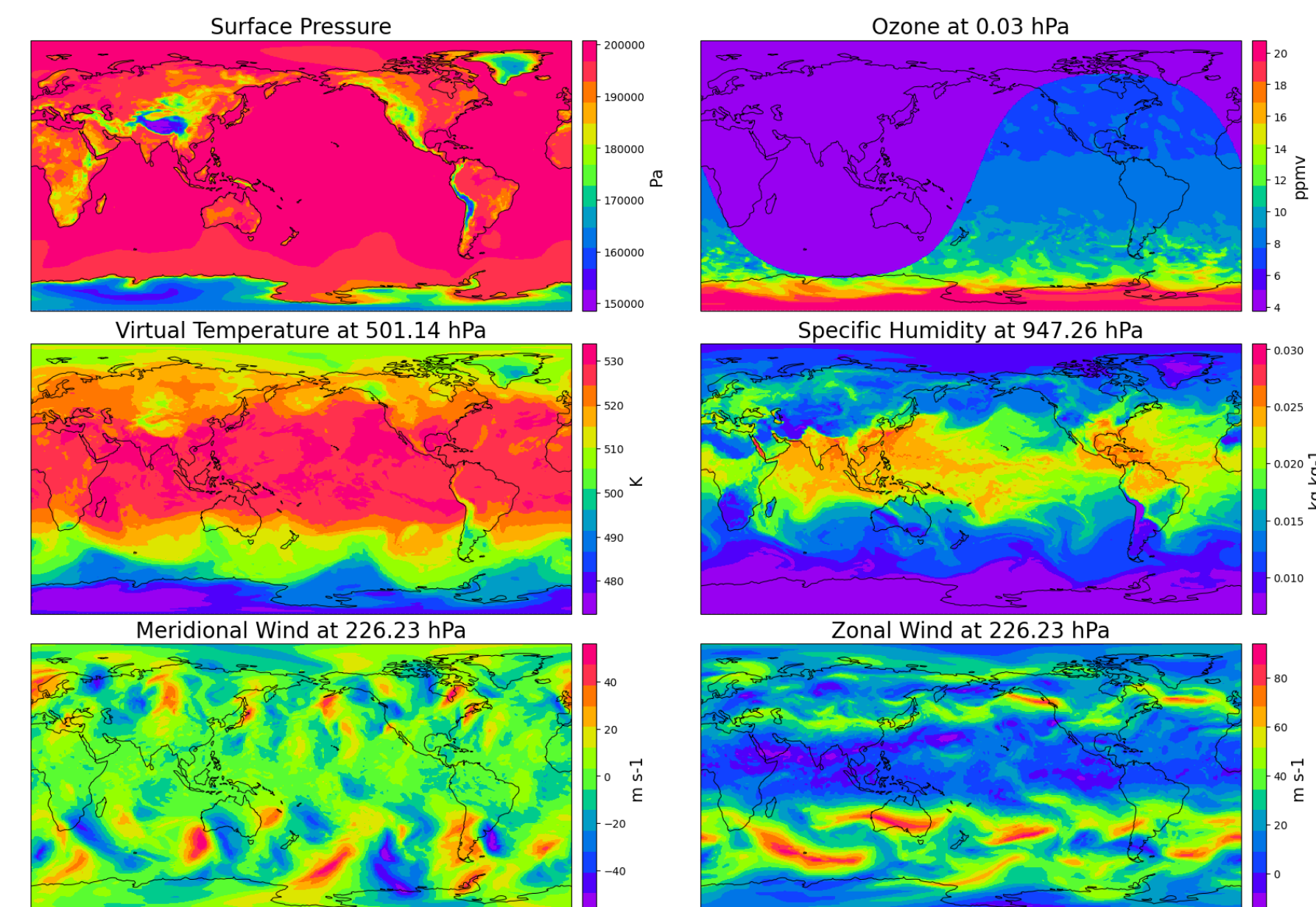


allowing for cheaply generated ensembles for DA.

DATA

NASA's GEOS-IT [1] data is a product of NASA's assimilation system for instrument teams.

Variables: Ozone, Specific Humidity, Meridional Wind, Zonal Wind, Virtual Temperature, Surface Pressure
Coordinates: Latitudes: 360, Longitudes: 576, Levels: 72
Temporal: 6 hourly Years 2000-2024
361 features



METHODOLOGY: DATA ASSIMILATION & DIFFUSION MODELING

DATA ASSIMILATION involves combining data from various modalities to construct an estimates of unknown states across time. NASA implements *hybrid 4D ensemble variational (hybrid 4D Ens-Var)* [2] data assimilation, which aims to minimize $\mathcal{J}$ w.r.t. $\delta \mathbf{x} = (\delta \mathbf{x}_0, \delta \mathbf{x}_1, \dots, \delta \mathbf{x}_T)$ for the times $[t_0, t_T]$

$$\mathcal{J}(\delta \mathbf{x}) = \underbrace{\frac{1}{2} \mathbf{x}^\top B^{-1} \delta \mathbf{x}}_{\text{initial condition loss}} + \underbrace{\frac{1}{2} (\mathbf{y} - \mathbf{y}_o)^\top R^{-1} (\mathbf{y} - \mathbf{y}_o)}_{\text{observation loss}} + \underbrace{J_c}_{\text{additional constraints (e.g. forecast model error)}}$$

$\mathbf{x}_b$: "background" $\mathbf{y} = H(\mathbf{x}_b + \delta \mathbf{x})$: prediction at obs.
 H : observation operator R : observation error covariance
 $B = \beta_s B_s + \beta_e B_e$: weighted background covariance matrix static ensemble

OUR GOAL: Estimate the ensemble component of the background error cov. B via an AI-based ensemble.

Construct $\mathbf{X} = \frac{1}{\sqrt{N-1}} [(\mathbf{x}_1 - \bar{\mathbf{x}}), (\mathbf{x}_2 - \bar{\mathbf{x}}), \dots, (\mathbf{x}_N - \bar{\mathbf{x}})]$ with N ensemble members such that $B_e = \mathbf{X} \mathbf{X}^\top$ from a diffusion model.

DIFFUSION MODELING [3] learns a mapping from a noisy, usually Gaussian distribution to a target distribution, thereby providing means for generating samples from the target distribution. This diffusion model allows us to generate samples to estimate B_e .

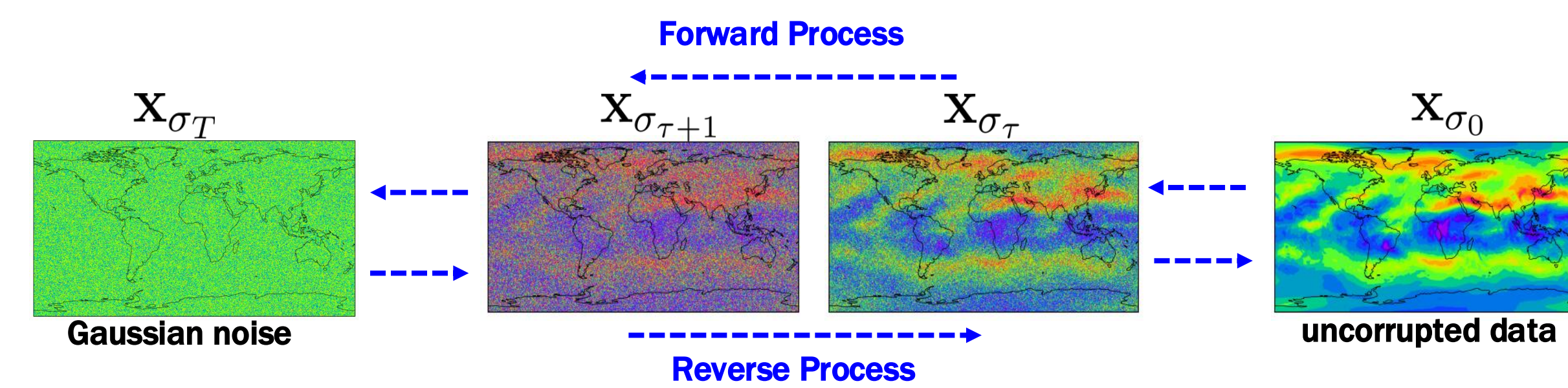
Forward Process: Corrupt with Gaussian noise

$$\mathbf{x}_{\sigma_\tau} = \mathbf{x}_{\sigma_0} + \sigma_\tau \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Reverse Process: Denoise to the target data

$$\frac{d\mathbf{x}_{\sigma_\tau}}{d\tau} = -\sigma_\tau^2 \nabla_{\mathbf{x}} \log p_\tau(\mathbf{x}_{\sigma_\tau}) d\tau + \sigma_\tau d\mathbf{w}_\tau$$

learnable



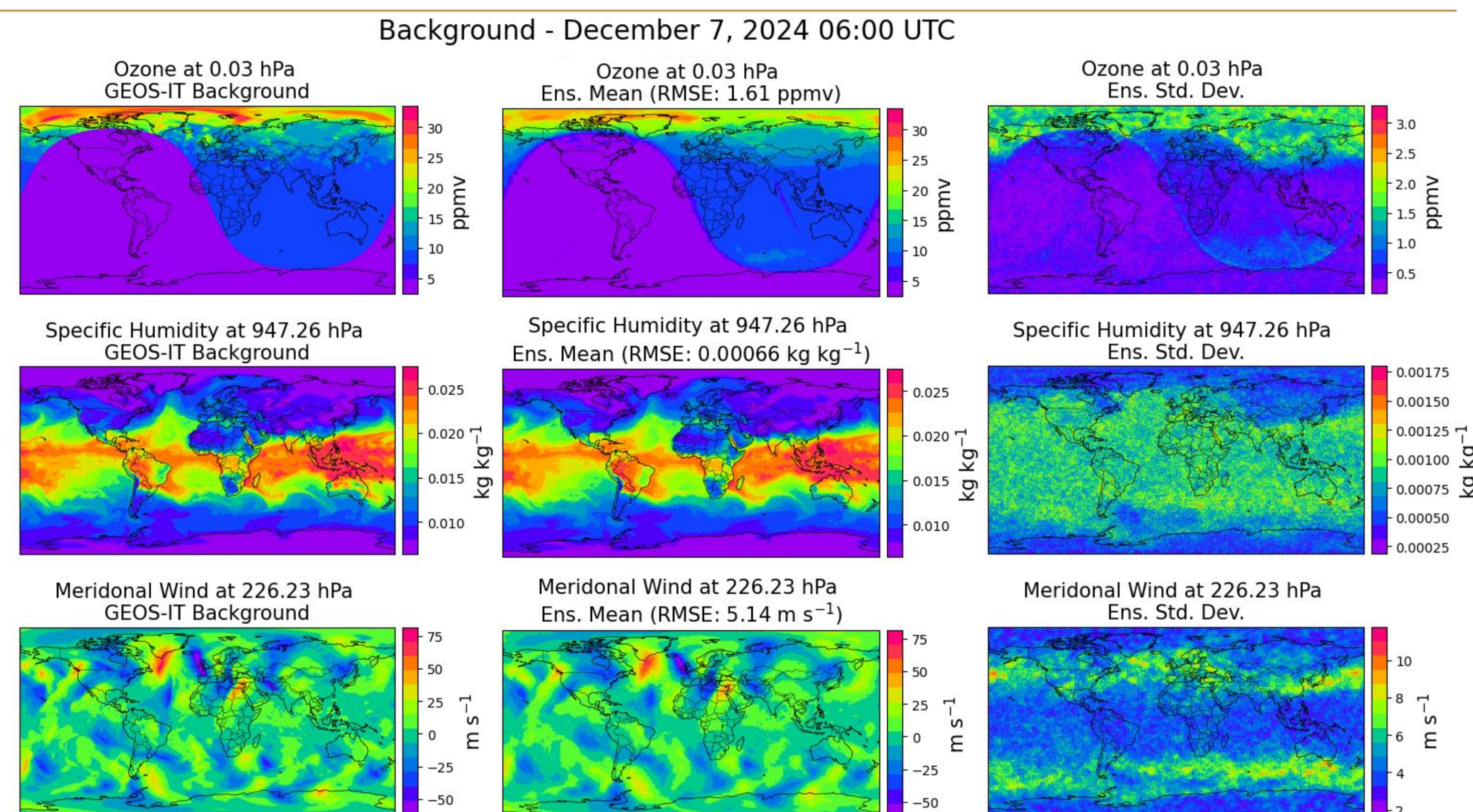
RESULTS

We train a SwinV2 Transformer [4] on a subset of 6 features.

(Left column) Target GEOS-IT background state.

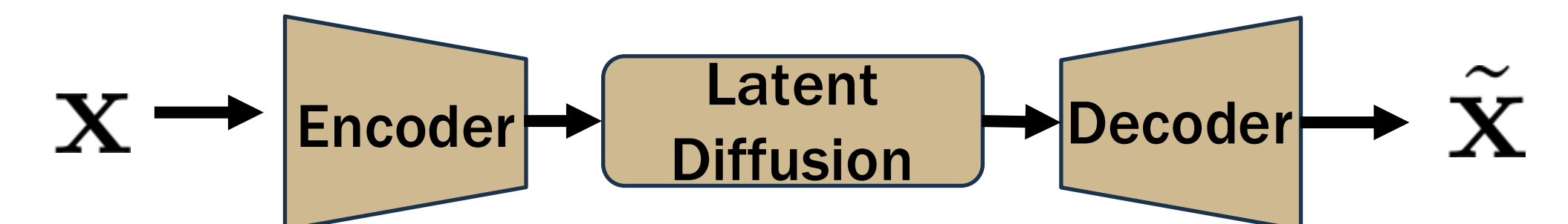
(Center column) ensemble mean prediction from $N=25$ ensemble members.

(Right column) ensemble mean standard deviation from $N=25$ ensemble members.



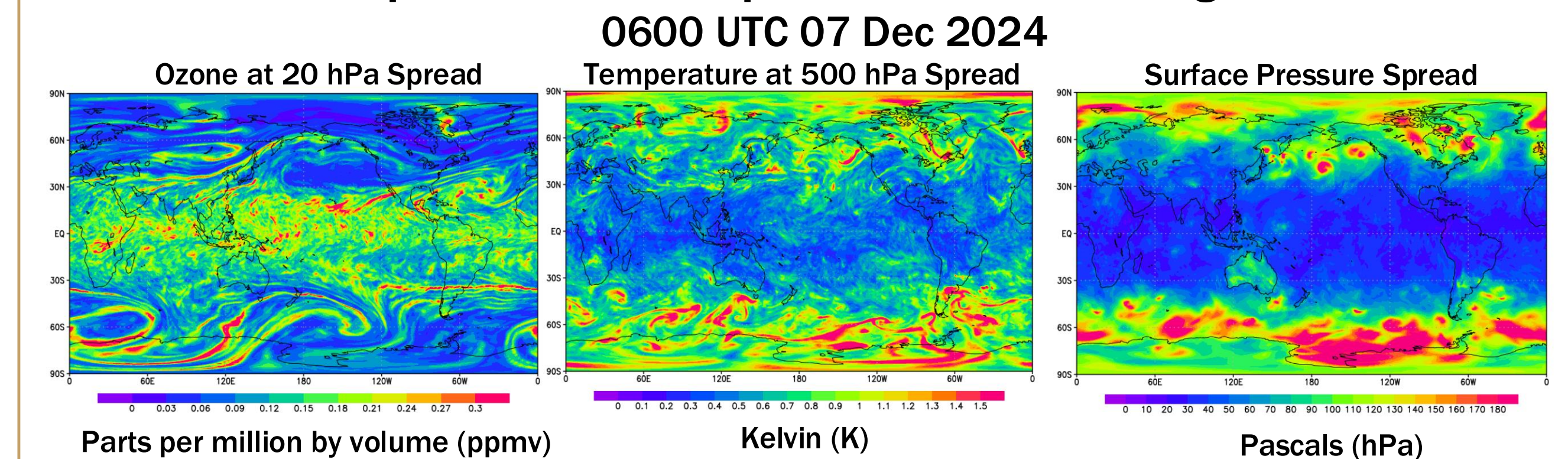
ONGOING DIRECTIONS

LATENT DIFFUSION MODELING [5] compresses the data to a lower dimensional latent space, allowing for cheaper computations and learning in this latent space.



VALIDATING ENSEMBLE SPREAD involves comparing the ensemble spread from the diffusion model vs. from an ensemble of outputs from the physics-based system (GEOS-FP [6]) to ensure physical consistency.

Examples of ensemble spread for GEOS-FP backgrounds:

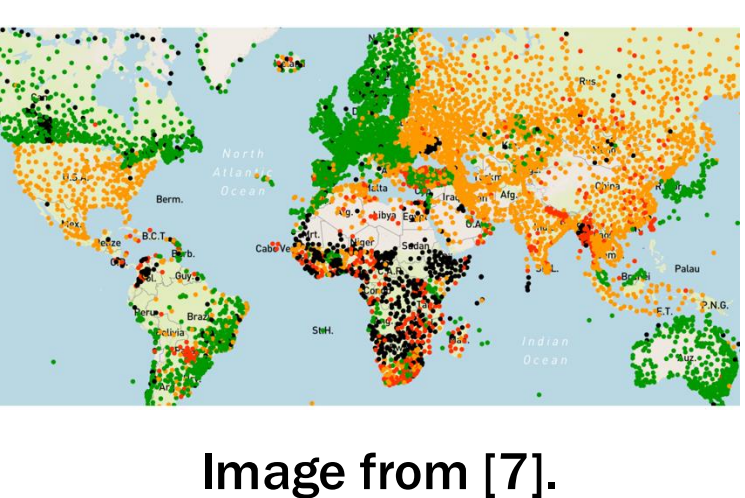


FUTURE DIRECTIONS

INTEGRATION INTO NASA'S GEOS HYBRID DATA ASSIMILATION SYSTEM. Prototype the integration of the AI-based ensemble into NASA's GEOS hybrid 4D-Ens-Var data assimilation scheme prior to operational use.



SENSOR PLACEMENT. The placements of sensors to collect observations can be optimized to maximize the information learned about the system. AI-based components accelerate this process.



REFERENCES

- [1] Global Modeling and Assimilation Office. "Goddard Earth Observing System (GEOS) for Instrument Teams." NASA Goddard Space Flight Center, gmao.gsfc.nasa.gov/gmao-products/geos-iv/.
- [2] Todling, Ricardo, and Amal El Akkroui. The GMAO Hybrid Ensemble-Variational Atmospheric Data Assimilation System: Version 2.0. NASA Technical Memorandum NASA/TM-2018-104606/Vol. 50, National Aeronautics and Space Administration, Goddard Space Flight Center, Mar. 2018. NASA Technical Reports Server, ntrs.nasa.gov/citations/20180002172.
- [3] Karras, Tero, et al. "Elucidating the Design Space of Diffusion-Based Generative Models." Advances in Neural Information Processing Systems 35 (NeurIPS 2022), edited by S. Koyejo et al., Curran Associates, 2022, pp. 26565-77.
- [4] Liu, Ze, et al. "Swin Transformer V2: Scaling Up Capacity and Resolution." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 12009-19. CVF Open Access, openaccess.thecvf.com/content/CVPR2022/papers/Liu_Swin_Transformer_V2_Scaling_Up_Capacity_and_Resolution_CVPR_2022_paper.pdf.
- [5] Rombach, Robin, et al. "High-Resolution Image Synthesis with Latent Diffusion Models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 10684-10695.
- [6] Global Modeling and Assimilation Office. "GEOS Near-Real Time Data Products." NASA Goddard Space Flight Center, gmao.gsfc.nasa.gov/gmao-products/geos-near-real-time-data-products/.
- [7] ECMWF Newsletter, no. 177, Autumn 2023, edited by Georg Lentze, European Centre for Medium-Range Weather Forecasts, www.ecmwf.int/en/elibrary/81513-newsletter-no-177-autumn-2023.